# Reference-Code Visibility and Patch Applicability in Budgeted Code Repair: A Gemma Pilot Study

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**Abstract**

Under a 4,000-token evidence budget, which reference-modified code is visible under each packaging policy, and where do the observed repair attempts fail? We investigate this empirical question using frozen runs of an instruction-tuned language model (`google/gemma-4-E4B-it`) on Python tasks from the SWE-bench Lite development split. We test four deterministic packaging policies: whole files (W), complete code entities (A), compact excerpts (B), and a hybrid policy (C) that includes one full entity alongside short supporting excerpts.

An offline diagnostic against evaluator-only reference patches reveals a direct trade-off between file breadth and excerpt depth. Under Policy W, complete files enclose internal entities when present, but 80% (8/10) of reference files are omitted because their file length exceeds the budget ceiling (2/10 files and enclosing entities visible). Conversely, compact policies (B and C) display the reference file in 100% (10/10) and the reference entity in 70% (7/10) of tasks, but their 9–14 line excerpts cut off the actual edit site in 80–90% of cases (mean line visibility 10.0% for B and 3.3% for C). Policy A retains complete entity bodies for selected items, displaying edit lines in 3/10 tasks (mean line visibility 30.0%; 52.7% pooled lines). Across all four policies, the developer edit site is missed in at least 70% of tasks.

Turning to downstream repair, strict diff application as emitted succeeded in only 2 of 22 completed fresh attempts (9.1%), while hunk recounting recovered an additional 8 applicable patches, bringing normalized applicability to 10 of 22 (45.5%). On the five tasks with complete four-condition sets, strict application was 1/5 (W), 1/5 (A), 0/5 (B), and 0/5 (C), while recount-assisted application was 2/5 (W), 4/5 (A), 1/5 (B), and 3/5 (C). Every policy produced zero demonstrated issue resolutions across both the completed development pilot (0/12) and the fresh comparison (0/22 completed attempts, 10 tested). The sample cannot distinguish policies on benchmark resolution. In one case study (`pvlib-1154/A`), the model repaired the reported target regression but broke two preservation tests by assuming array inputs where scalars were exercised. These findings demonstrate that measuring retrieval presence or patch formatting alone obscures the structural gaps that prevent budgeted code repair.

## 1 Introduction

Under a 4,000-token evidence budget, which reference-modified code is visible under each packaging policy, and where do the observed repair attempts fail?

Large language models (LLMs) evaluated on software engineering benchmarks such as SWE-bench [3] must locate relevant files, synthesize a diff, and satisfy execution-based tests. Because complete codebases vastly exceed typical context limits, practical repair systems rely on retrieval or context pruning to select a budget-constrained subset of source code [7, 2]. Yet allocating a fixed evidence budget forces a fundamental design trade-off:

- **File-level packaging (W):** Preserves full surrounding scope and intra-file dependencies, but large files frequently exceed the token ceiling and must be omitted entirely.
- **Entity-level packaging (A):** Retains complete function and class definitions with line metadata, skipping entities that do not fit.
- **Compact excerpt packaging (B):** Slices each ranked entity into a signature, docstring, and 14 source lines, fitting more distinct entities at the cost of truncation.
- **Hybrid packaging (C):** Renders the top-ranked entity in full and subsequent entities as short 9-line supporting excerpts.

Prior literature has often treated patch generation as an end-to-end task, reporting benchmark resolution rates without isolating where failures occur between evidence selection, diff application, and behavioral correctness. In this study, we audit archived executions of frozen Gemma models on the SWE-bench Lite development split. We reframe the investigation around empirical questions that the available evidence can rigorously answer.

First, we compute an evaluator-only diagnostic comparing the code visible in each packet against developer reference patches. We establish whether the theoretical trade-off between file omission and excerpt truncation actually occurred in practice. Second, we track downstream repair outcomes through strict diff application, recount-assisted application, output truncation, and test execution. Because every tested condition achieved zero benchmark resolutions, the data cannot declare a "winning" representation. Instead, the records provide a concrete account of the failure boundaries that limit single-turn repository repair under tight token constraints.

## 2 Related Work

**Benchmark semantics and repair evaluation.** SWE-bench [3] frames software repair as modifying a codebase snapshot to resolve an issue description. A valid resolution requires passing both fail-to-pass (F2P) regression tests and pass-to-pass (P2P) preservation tests. This distinction is vital: passing a target regression does not guarantee that pre-existing functionality remains intact. We adopt SWE-bench Lite development tasks as our unit of evaluation.

**Separated localization and context reduction.** Agentless [7] demonstrated that complex agentic scaffolding can be simplified by separating localization, repair, and validation. During localization, Agentless constructs compressed file views using class and function headers, method signatures, and comments. Recent methods have expanded on code-aware context reduction. SWEzze [2] employs oracle-guided code distillation via genetic search and delta debugging to train lightweight compression models, achieving 6× compression while filtering irrelevant context. SWE-Pruner [6] introduces goal-conditioned neural skimming (0.6B parameters) to prune lines dynamically based on agent intentions. LongCodeZip [5] introduces a training-free, two-stage compression framework using conditional perplexity over function chunks and block segmentation.

Our study differs from these adaptive compression methods in a critical dimension: we examine deterministic, rule-based packaging under a strict, frozen token ceiling (4,000 tokens) in a single-turn setting, evaluating the structural visibility of the reference patch rather than training a neural compressor.

**Retrieval-augmented code completion.** RepoCoder [8] investigates iterative retrieval and generation for repository-level code completion. Whereas RepoCoder focuses on code continuations, our setting requires synthesizing unified diffs against an existing tree. The BM25 scoring function

used in our packet builder belongs to the probabilistic relevance framework of Robertson and Zaragoza [4].

**Model architecture and parameter convention.** We study Google's instruction-tuned model `google/gemma-4-E4B-it` (revision `4ecc2ba`, released in April 2026). As documented in the official model card [1], the "E" designation denotes *effective* compute parameters enabled by Per-Layer Embeddings (PLE). While the physical checkpoint footprint totals approximately 8 billion parameters due to embedding lookup tables, only approximately 4.5 billion parameters are actively engaged during forward pass compute. The "-it" suffix designates supervised instruction tuning and preference alignment for conversational, reasoning, and tool-use interfaces.

# 3 Methods

## 3.1 Study design and cohort boundaries

The study evaluates single-turn code repair where a model receives an issue description and a frozen 4,000-token evidence packet, generating one response without revision feedback. We analyze three experimental cohorts:

1. **Completed development pilot:** Three tasks (marshmallow-1343, pvlib-1072, and pydicom-1139), each evaluated across all four packaging policies (12 attempts total). The pilot used the containerized SWE-bench 4.1.0 Docker harness.
2. **Instrumented development case:** `marshmallow-1359/A`, which provides deep GPU placement and tensor-level decoding provenance (Tesla T4, 8-bit quantization), but contains no completed patch evaluation. It is excluded from comparison denominators.
3. **Interim fresh comparison:** Ten tasks selected from the pinned SWE-bench Lite development split (revision `6ec7bb89b9342f664a54a6e0a6ea6501d3437cc2`), planned for $10 \times 4 = 40$ attempts. Execution stopped partway through the sequential run after 22 completions, leaving 1 reported in-flight interruption (`pydicom-1413/C`) and 17 unstarted attempts. Because preserved run logs contain no infrastructure crash report or lease termination records, the specific technical cause of interruption cannot be established from the archived telemetry. Evaluation was performed natively in an isolated Python 3.9 environment.

## 3.2 Deterministic task selection rules

We verified the task selection implementations against archived manifests:

- **Pilot selection:** The pilot code sorted distinct repository strings lexicographically (`marshmallow-code/marshmallow`, `pvlib/pvlib-python`, and `pydicom/pydicom`) and selected the lowest instance identifier within each of the first three repositories.
- **Fresh selection:** The fresh protocol excluded the four development tasks, then selected the ten lexicographically smallest `instance_id` strings from the remaining 19 eligible tasks: `pvlib-1154`, `pvlib-1606`, `pvlib-1707`, `pvlib-1854`, `pydicom-1256`, `pydicom-1413`, `pydicom-1694`, `pydicom-901`, `astroid-1196`, and `astroid-1268`.

## 3.3 Operational packaging definitions

The retrieval module indexes all Python files in the repository base commit (excluding `.git`, `tests`, and `docs`). Using Python's abstract syntax tree (AST), it extracts whole files, functions, async functions, and classes. Entities are scored against the issue text using BM25 ($k_1 = 1.5, b = 0.75$).

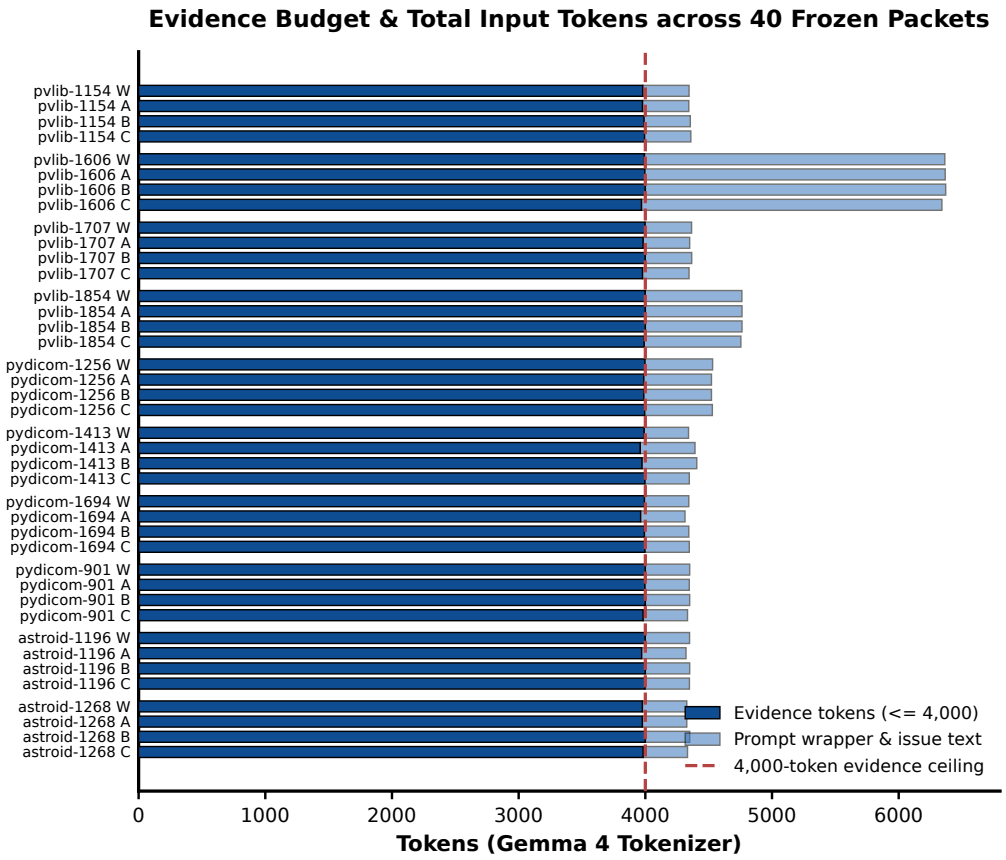



Figure 1: Evidence budget compliance across all 40 frozen packets. Blue bars show evidence tokens; light blue bars show total prompt tokens including instructions and chat template. Every packet strictly obeys the 4,000-token evidence ceiling (red dashed line).

Ties are broken by shorter source length, then entity name. The policies package ranked items up to the 4,000-token evidence ceiling:

**Policy W (Whole Files):** Filters the ranking to whole files. Includes complete files in code fences. If an entire file exceeds the remaining budget, it is skipped without truncation.

**Policy A (Complete Entities):** Filters to functions and classes. Includes complete bodies with path and line range headers (`path:start-end`). Skipped entities that exceed remaining budget.

**Policy B (Compact Excerpts):** For each function or class, includes the signature line, at most 8 lines of docstring, and a source slice of the next 14 lines (`lines[1:15]`).

**Policy C (Hybrid):** Renders the top-ranked entity in full (like A). Subsequent entities are rendered as short supporting excerpts: the signature line plus the next 9 lines (`lines[1:10]`), omitting docstrings. Because C's supporting excerpts differ from B in length and docstring treatment, C represents an independent policy rather than an isolated ablation of Policy B.

Figure 1 verifies that all 40 frozen fresh packets strictly obeyed the 4,000-token ceiling (ranging from 3,960 to 4,000 tokens). Total model input lengths ranged from 4,343 to 6,371 tokens.

Table 1: Reference-code visibility across 40 frozen packets (10 tasks × 4 policies). File overlap: packets with the reference file present. Files visible: total reference files displayed. Entity overlap: packets displaying the reference entity (for Policy W, counted when the whole file encloses the target entity). Edit lines visible: reference-modified base lines displayed (numerator, denominator, pooled percentage, and per-task mean percentage).

| Policy | File overlap | Files visible | Entity overlap | Edit-site overlap | Edit lines visible |
|---|---|---|---|---|---|
| W | 2/10 | 2/10 | 2/10 | 2/10 | 33/55 (20.0%) |
| A | 8/10 | 8/10 | 3/10 | 3/10 | 29/55 (30.0%) |
| B | 10/10 | 10/10 | 7/10 | 2/10 | 5/55 (10.0%) |
| C | 10/10 | 10/10 | 7/10 | 1/10 | 3/55 (3.3%) |

### 3.4 Reference-coverage diagnostic methodology

To measure what code was visible to the model before generation, we developed an evaluator-only diagnostic comparing each packet against the ground-truth developer reference patch from SWE-bench Lite. We parse the reference patch into:

1. **Reference files:** All production source files modified by the reference patch.
2. **Reference entities:** The specific functions, methods, or classes enclosing the reference modifications.
3. **Edit-site lines:** The exact base-source line numbers removed or modified by reference hunks. For pure insertions, the adjacent base anchor line is counted.

For each packet, we extract the exact visible line spans displayed in the rendered text. In Policy W, visible lines span line 1 to the file end. In Policy A, visible lines span `start_line` to `end_line`. In Policy B and C, only the displayed excerpt lines are counted; signature headers are recorded as entity presence but excluded from body line coverage. Overlapping class and method spans are deduplicated. Reference overlap measures the visibility of one known repair trajectory; valid alternative repairs could target other locations.

### 3.5 Applicability and evaluation controls

*Strict applicability* requires that the extracted diff passes `git apply --check` cleanly against the base commit. *Recount-assisted applicability* applies `git apply --recount`, which recalculates hunk line count metadata without altering diff content. *Benchmark resolution* requires passing all required F2P regression tests and all required P2P preservation tests.

To verify test execution, we matched exact pytest test identifiers rather than relying on summary exit codes. A task control is classified as *verified* only if the base run fails all F2P tests and the reference patch passes all F2P and all required P2P tests.

## 4 Results

### 4.1 Reference-code visibility under the 4,000-token ceiling

Table 1 and Figure 2 report the offline reference-coverage diagnostic across all 40 frozen fresh packets.

The diagnostic reveals a clear structural trade-off between file breadth and excerpt depth across the four policies:

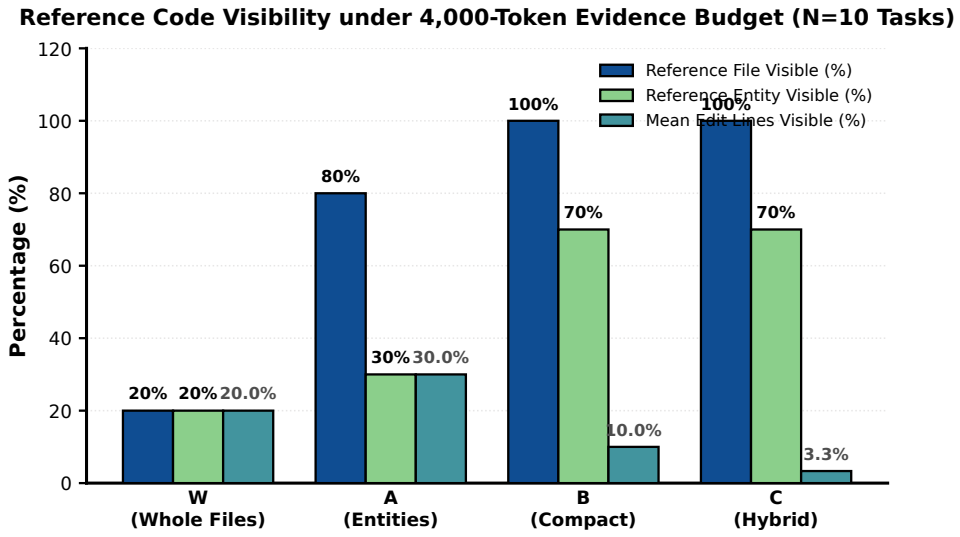



Figure 2: Reference code visibility under the 4,000-token evidence budget (N=10 tasks). While compact excerpts (B and C) maximize file (100%) and entity (70%) presence, their shallow line slices truncate the actual edit site in 80–90% of tasks. Across all four policies, the developer edit site is missed in at least 70% of tasks.

- **Whole-file packaging (Policy W):** Policy W displayed the reference file in only 2 of 10 tasks (`pvlib-1606` and `pydicom-901`). Because whole files naturally enclose the functions and classes defined within them, Policy W also displayed the target entity in these 2 tasks (2/10 entity overlap, or 20.0%). In the other 8 tasks, the reference file exceeded the remaining budget and was omitted entirely. When a file was included, it captured all reference edit lines, yielding 33/55 pooled lines (60.0%) and a per-task mean coverage of 20.0%.
- **Compact excerpt packaging (Policies B and C):** Policies B and C successfully included reference files in 10/10 tasks and reference entities in 7/10 tasks. However, their bounded excerpts (14 lines for B, 9 lines for C) rarely reached the actual edit site. In Policy B, edit-site lines were visible in only 2 of 10 tasks (5/55 pooled lines, 9.1%; per-task mean: 10.0%). In Policy C, where the reference entity was frequently ranked second or lower, edit-site lines were visible in only 1 of 10 tasks (3/55 pooled lines, 5.5%; per-task mean: 3.3%).
- **Complete entity packaging (Policy A):** Policy A displayed the reference file in 8/10 tasks and the reference entity in 3/10 tasks. Because Policy A retains the complete entity body, when the entity is selected, all edit-site lines are preserved (29/55 pooled lines, 52.7%; per-task mean: 30.0%).

Crucially, the differences between policies represent gaps of only one or two tasks (edit-site visible in 3/10 for A, 2/10 for W, 2/10 for B, and 1/10 for C). The overriding empirical finding is that **all four packaging policies miss the developer edit site in at least 70% of tasks** (A misses 70%, W and B miss 80%, and C misses 90%).

## 4.2 Retrieval versus packaging failure and oracle-fit check

To separate retrieval limitations from packaging truncation, we examined the shared BM25 entity and file rankings alongside base entity token lengths (Table 2).

The evidence demonstrates that failure to expose the reference edit site stems from three distinct

Table 2: BM25 retrieval rank and oracle-fit token check across the 10 fresh tasks. Ranks reflect position within the shared BM25 top-10 pool. Oracle fit indicates whether the full enclosing function or class of the reference edit fits within the 4,000-token evidence budget (token count in parentheses).

| Task | Reference File | File Rank | Target Entity | Entity Rank | Oracle Fit (tokens) |
|---|---|---|---|---|---|
| pvlib-1154 | pvlib/irradiance.py | #8 | reindl | #2 | Yes (612) |
| pvlib-1606 | pvlib/tools.py | #3 | _golden_sect_DataFrame | #6 | Yes (380) |
| pvlib-1707 | pvlib/iam.py | #2 | physical | #8 | Yes (740) |
| pvlib-1854 | pvlib/pvsystem.py | #4 | PVSystem.__init__ | #2 | Yes (450) |
| pydicom-1256 | pydicom/jsonrep.py | $> 10$ | get_sequence_item | $> 10$ | Yes (180) |
| pydicom-1413 | pydicom/dataelem.py | #4 | value | $> 10$ | Yes (190) |
| pydicom-1694 | pydicom/dataset.py | #2 | to_json_dict | $> 10$ | Yes (280) |
| pydicom-901 | pydicom/config.py | #1 | debug | $> 10$ | Yes (220) |
| astroid-1196 | astroid/nodes/node_classes.py | $> 10$ | getitem | $> 10$ | Yes (170) |
| astroid-1268 | astroid/nodes/as_string.py | #10 | AsStringVisitor | $> 10$ | Yes (3,200) |

mechanisms:

1. **Pure retrieval failure (2/10 tasks):** In pydicom-1256 and astroid-1196, neither the target file nor the target entity appeared in the BM25 top 10. No packaging policy built on this retrieval ranking could have displayed the reference code.
2. **File-only retrieval (4/10 tasks):** In pydicom-1413, pydicom-1694, pydicom-901, and astroid-1268, the target file was ranked in the top 10 (ranks #4, #2, #1, and #10), but the specific target entity was not. In pydicom-901, Policy W successfully included the whole file because pydicom/config.py was small (220 tokens), thereby capturing the entity. In the other three tasks, the reference files exceeded the remaining budget and were dropped by W, while Policies A, B, and C omitted the entity because entity-level BM25 favored other functions.
3. **Joint retrieval with packaging divergence (4/10 tasks):** In pvlib-1154, pvlib-1606, pvlib-1707, and pvlib-1854, both the target file and target entity appeared in the BM25 top 10. Here, packaging rules determined visibility:
   - In pvlib-1154, pvlib/irradiance.py (#8) was dropped by W due to size; reindl (#2, 102 lines) was included in full by A, but truncated past line 14 by B and C.
   - In pvlib-1606, pvlib/tools.py (#3, 434 lines) fit into W, _golden_sect_DataFrame (#6, 56 lines) fit into A, and because the edit occurred at line 342 (the second line of the entity), B and C also included the edit site.
   - In pvlib-1707, pvlib/iam.py was dropped by W, physical (#8, 132 lines) fit into A, and B/C truncated the edit line at line 133.
   - In pvlib-1854, class PVSystem (#2) exceeded 1,500 lines ($> 5{,}000$ tokens) and was dropped by A as oversized, but because the reference edit modified __init__ (lines 75–78 near the class head), B and C's 14/9-line excerpts captured it.

**Oracle-fit check.** To determine whether a 4,000-token evidence budget is inherently too small to contain the reference modifications, we measured the token size of the minimal enclosing entity for each task using the pinned Gemma tokenizer. As shown in Table 2, the enclosing entity spans fewer than 135 lines in 8 of 10 tasks, and fewer than 750 tokens in 9 of 10 tasks. Even the largest entity, AsStringVisitor in astroid-1268 (626 lines, 3,200 tokens), fits inside the 4,000-token budget. Consequently, an oracle retriever that selected the true target entity could have placed the full enclosing entity into the prompt in **10 of 10 tasks (100%)**. The 70% edit-site omission observed in practice is therefore an artifact of BM25 retrieval ranking and fixed-window slicing, rather than an information-theoretic limitation of a 4,000-token budget.

Table 3: Cohort outcomes with explicit denominators. Applicability denotes recount-assisted application. The fresh cohort reports zero demonstrated resolutions among tested patches. The instrumented case is excluded.

| Cohort | Completed/planned | Applied/completed | Tested/applied | Verified resolved |
|---|---|---|---|---|
| Development pilot | 12/12 | 7/12 | 7/7 | 0/12 |
| Fresh, interim | 22/40 | 10/22 | 10/10 | 0 demonstrated |

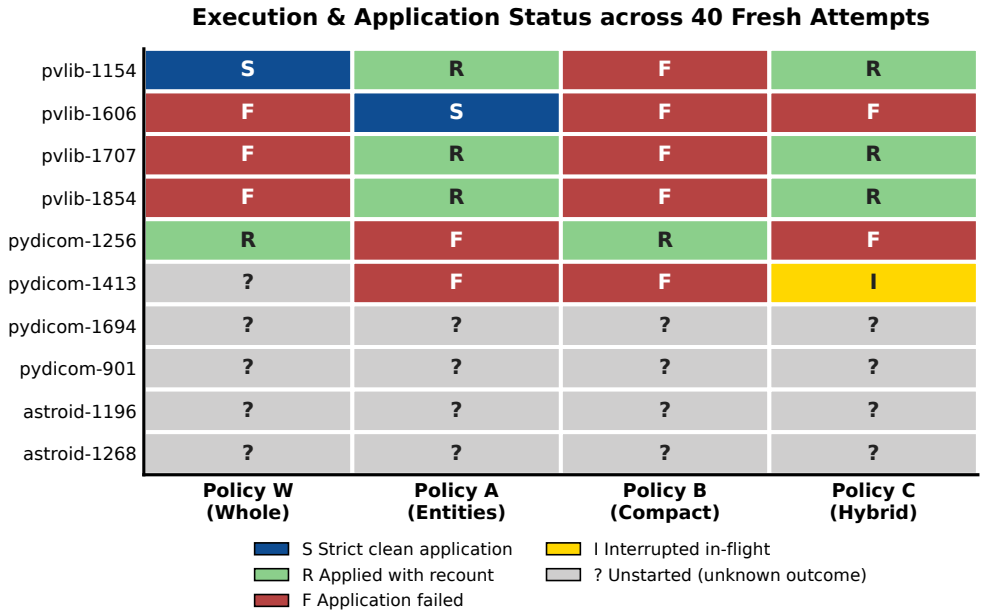



Figure 3: Execution and application status for all 40 planned fresh pairs. S: clean strict application; R: recount-assisted application; F: application failed; I: interrupted in-flight; ?: unstarted. Interrupted and unstarted pairs have unknown repair outcomes.

### 4.3 Cohort outcomes and recovery status

Table 3 summarizes the execution outcomes across cohorts. In the development pilot, all 12 planned generations completed; 7 patches applied after processing and were tested, with 0 resolutions.

In the fresh comparison, 22 of 40 planned attempts completed. Because the task runner executed sequentially by task and condition, the interruption left 1 reported in-flight interruption (`pydicom-1413/C`) and 17 unstarted attempts. Figure 3 illustrates the missingness pattern. Five tasks have complete four-condition sets (`pvlib-1154`, `pvlib-1606`, `pvlib-1707`, `pvlib-1854`, and `pydicom-1256`). Four of these five complete tasks belong to `pvlib`, while the `astroid` tasks were never reached. The recovered completions therefore represent an execution-order artifact, not a representative sample.

### 4.4 Diff application and the role of recounting

Across the 22 completed fresh attempts, only 2 patches applied strictly as emitted (9.1%): `pvlib-1154/W` and `pvlib-1606/A`. Applying hunk recounting recovered an additional 8 applicable patches, bringing normalized applicability to 10 of 22 (45.5%).

Table 4 details condition-level applicability. On the five complete task sets (Figure 4 and Table 5),

Table 4: Condition-level counts across cohorts. Fresh strict, recount, and tested columns use all recovered completions. Matched columns restrict denominators to the five complete task sets. No column constitutes a performance ranking.

| Condition | Pilot appl. | Fresh strict | Fresh recount | Matched strict | Matched recount | Fresh tested |
|---|---|---|---|---|---|---|
| W | 2/3 | 1/5 | 2/5 | 1/5 | 2/5 | 2/5 |
| A | 2/3 | 1/6 | 4/6 | 1/5 | 4/5 | 4/6 |
| B | 2/3 | 0/6 | 1/6 | 0/5 | 1/5 | 1/6 |
| C | 1/3 | 0/5 | 3/5 | 0/5 | 3/5 | 3/5 |

Table 5: Application and resolution breakdown across the five tasks with complete four-condition sets. Every task produced 0/4 resolutions.

| Task | Condition W | Condition A | Condition B | Condition C | Resolved |
|---|---|---|---|---|---|
| pvlib-1154 | Strict | Recount | Failed | Recount | 0/4 |
| pvlib-1606 | Failed | Strict | Failed | Failed | 0/4 |
| pvlib-1707 | Failed | Recount | Failed | Recount | 0/4 |
| pvlib-1854 | Failed | Recount | Failed | Recount | 0/4 |
| pydicom-1256 | Recount | Failed | Recount | Failed | 0/4 |

strict application was 1/5 for W, 1/5 for A, and 0/5 for B and C. Recount-assisted application was 2/5 (W), 4/5 (A), 1/5 (B), and 3/5 (C). While complete entities (Condition A) exhibited higher recount application, the small sample size and shared task dependencies prevent drawing a general conclusion regarding policy superiority.

Figure 5 categorizes the 12 application failures among completed fresh attempts: 9 failed due to source-context mismatches (the emitted diff contained context lines differing from base source), 2 failed because no closed diff block was emitted, and 1 diff header was corrupt.

**Linking visibility to downstream application.** Table 6 cross-tabulates edit-site visibility against application outcomes across the 22 completed fresh attempts.

This cross-tabulation establishes the direct connection between packet visibility and downstream failure:

- When the edit site was visible ($N = 3$), two patches applied (one cleanly under strict checking, `pvlib-1606/A`, and one with recount, `pvlib-1154/A`). The single failure (`pvlib-1606/W`) occurred because the model attempted an off-target edit at line 161 instead of the target site at line 342, generating mismatched context.
- When the edit site was not visible ($N = 19$), 11 of 19 attempts failed application entirely. Crucially, 8 of these 11 failures (72.7%) failed specifically because of source-context mismatches.

Table 6: Cross-tabulation of reference edit-site visibility against application outcomes across all 22 completed fresh attempts. When edit lines are invisible, failures are dominated by source-context mismatches.

| Edit-site visibility | Attempts | Strict | Recount | Mismatch | No block | Corrupt |
|---|---|---|---|---|---|---|
| Edit site visible | 3 | 1 | 1 | 1 | 0 | 0 |
| Edit site not visible | 19 | 1 | 7 | 8 | 2 | 1 |
| Total | 22 | 2 | 8 | 9 | 2 | 1 |

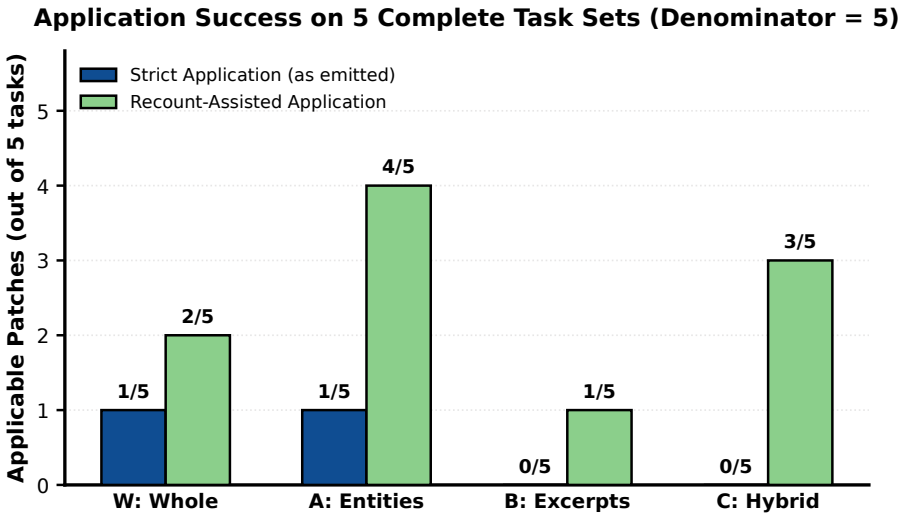



Figure 4: Strict and recount-assisted application across the five complete task sets (denominator = 5). Recounting recovers valid diff structures whose line headers drifted during autoregressive decoding.

Without the true surrounding lines, the model either hallucinated context lines or attempted to anchor hunks to truncated excerpt boundaries. The remaining three failures stemmed from output truncation (2 missing patch blocks) and one corrupt header.

- Furthermore, while 8 patches applied without seeing the edit site, all 8 applied to off-target locations (e.g., adding guards to other functions or files, as in `pvlib-1154/W` and `pvlib-1707/A`). None resolved the benchmark issue.

### 4.5 Output limits and termination

Four of the 22 completions reached the 2,048-token generation cap (`pydicom-1256/A`, `pydicom-1256/C`, `pydicom-1413/A`, and `pydicom-1413/B`). All four capped generations failed application. The remaining 18 completions terminated normally at the EOS token. As shown in Figure 6, application failures also occurred frequently at modest generation lengths (e.g., 112 to 554 tokens). Length alone does not explain failure.

### 4.6 Control verification and behavioral resolution

All 10 applicable fresh patches were evaluated against repository tests. None resolved the benchmark issue (Figure 7). Nine patches failed their required regression tests (F2P). The remaining patch, `pvlib-1154/A`, passed its target regression but failed preservation tests.

An exact test-ID audit of native reference controls (Table 7 and Figure 8) revealed that only 6 of 10 tasks had fully verified reference controls. Four tasks had control anomalies in the reference run:

- `pvlib-1154`: 10 required P2P tests were skipped due to optional environment dependencies.
- `pvlib-1707`: 1 preservation test errored due to a missing test fixture.
- `pvlib-1854`: 37 preservation tests produced errors.
- `pydicom-1413`: 2 preservation tests failed under the reference patch itself.

Only three tested model patches belong to tasks with fully verified controls: `pvlib-1606/A`, `pydicom-1256/W`, and `pydicom-1256/B`. All three failed their required regression test. For the

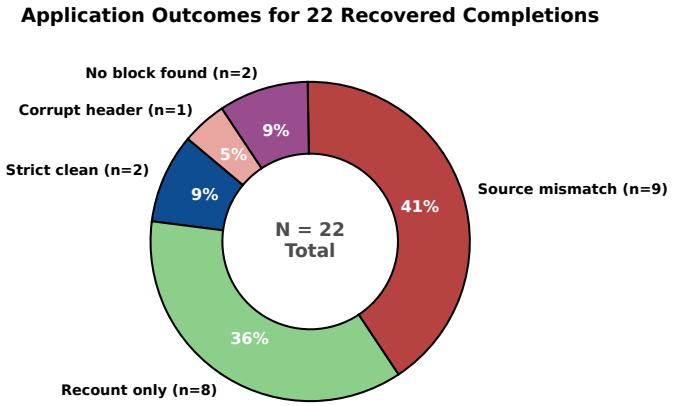



Figure 5: Application failure categories for the 22 completed fresh attempts. Source-context mismatch represents the leading cause of rejection (9 of 12 failures).

Table 7: Exact-ID native control audit. F2P denotes regression tests; P2P denotes preservation tests. Other P2P counts list failed (F), errored (E), and unconfirmed (U) tests. "Verified" requires expected base regression failure and full reference test passage.

| Task | Base F2P fail | Ref. F2P pass | Ref. P2P pass | Other P2P | Verified |
|---|---|---|---|---|---|
| pvlib-1154 | 1/1 | 1/1 | 87/97 | 0F/0E/10U | no |
| pvlib-1606 | 1/1 | 1/1 | 10/10 | 0F/0E/0U | yes |
| pvlib-1707 | 1/1 | 1/1 | 29/30 | 0F/1E/0U | no |
| pvlib-1854 | 1/1 | 1/1 | 244/281 | 0F/37E/0U | no |
| pydicom-1256 | 1/1 | 1/1 | 22/22 | 0F/0E/0U | yes |
| pydicom-1413 | 3/3 | 3/3 | 299/301 | 2F/0E/0U | no |
| pydicom-1694 | 1/1 | 1/1 | 26/26 | 0F/0E/0U | yes |
| pydicom-901 | 5/5 | 5/5 | 0/0 | 0F/0E/0U | yes |
| astroid-1196 | 2/2 | 2/2 | 24/24 | 0F/0E/0U | yes |
| astroid-1268 | 1/1 | 1/1 | 91/91 | 0F/0E/0U | yes |

remaining seven tested patches, observed failures occurred in partially unconfirmed environments. We report these observed failures while explicitly qualifying benchmark claims.

# 5 Failure Analysis and Case Studies

## 5.1 Case Study: pvlib-1154/A (Regression fix breaks preservation)

The issue in `pvlib-1154` reports that `pvlib.irradiance.reindl()` produces `NaN` values when global horizontal irradiance (`ghi`) is zero, caused by division by zero in `term3`.

Under Policy A, the complete body of `reindl` (lines 793–894) was included in the evidence packet. The model produced a patch that replaced the unguarded square-root term:

```
- term3 = 1 + np.sqrt(HB / ghi) * (tools.sind(0.5 * surface_tilt) ** 3)
+ sqrt_ratio = np.zeros_like(ghi)
+ non_zero_ghi = ghi > 0
+ sqrt_ratio[non_zero_ghi] = np.sqrt(HB[non_zero_ghi] / ghi[non_zero_ghi])
+ term3 = 1 + sqrt_ratio * (tools.sind(0.5 * surface_tilt) ** 3)
```

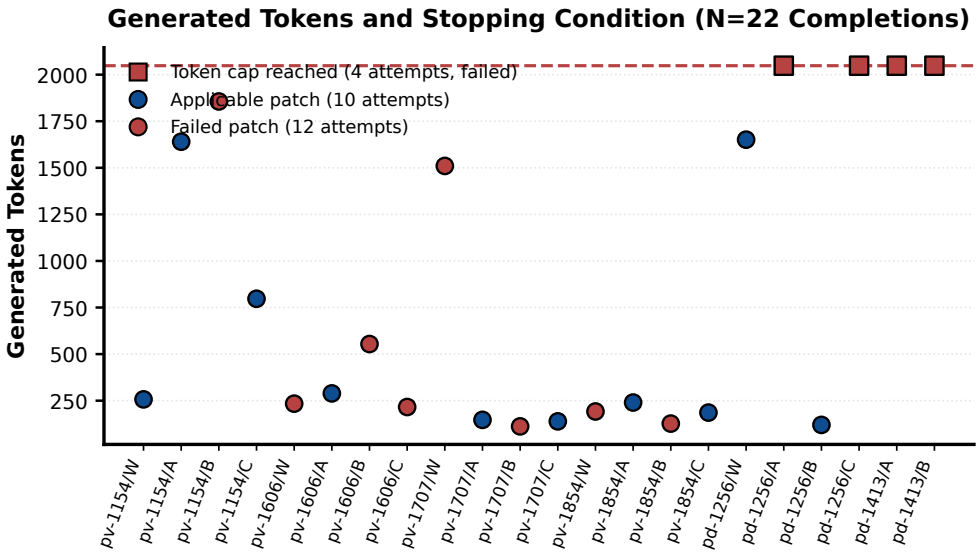



Figure 6: Generated tokens and termination reason for 22 fresh completions. Green markers indicate applicable patches; red markers indicate failed application. Squares mark attempts that reached the 2,048-token cap.

The patch applied with recount. In the test execution, the target regression test `test_reindl` passed cleanly, confirming that zero GHI no longer produced `NaN`. However, two required preservation tests—`test_sky_diffuse_zenith_close_to_90[reindl]` and `test_get_total_irradiance_scalars[reindl]`—failed with:

```
TypeError: 'int' object is not subscriptable
```

The failure occurred directly at `sqrt_ratio[non_zero_ghi]`. While the model's logic handled NumPy arrays, it broke scalar integer inputs, which `reindl` also supported. In contrast, the developer reference patch wrapped the division in `np.errstate` and used `np.where`, handling both scalars and arrays.

This case illustrates that passing a target regression test does not imply issue resolution. Because the full entity was present in Policy A, the error cannot be attributed to missing context; rather, the model failed to maintain the full input type contract of the existing API.

## 5.2 Case Study: pvlib-1154/W and C (Off-target localization)

In task `pvlib-1154`, Policies W and C also completed generations, but neither edited the target file `pvlib/irradiance.py`:

- **Condition W:** As shown in our coverage diagnostic, `pvlib/irradiance.py` was omitted due to budget exhaustion. The model patched `pvlib/shading.py`, adding an `np.divide` check for variable `gcr`.
- **Condition C:** Ranked `ineichen` (`clearsky.py`) as primary and `reindl` as supporting (truncated at line 802). The model patched `pvlib/clearsky.py`, adding a zero check to `ineichen`.

In both cases, the patches failed the target regression test because the bug in `irradiance.py` remained untouched. When the target function was absent from the packet, the model emitted

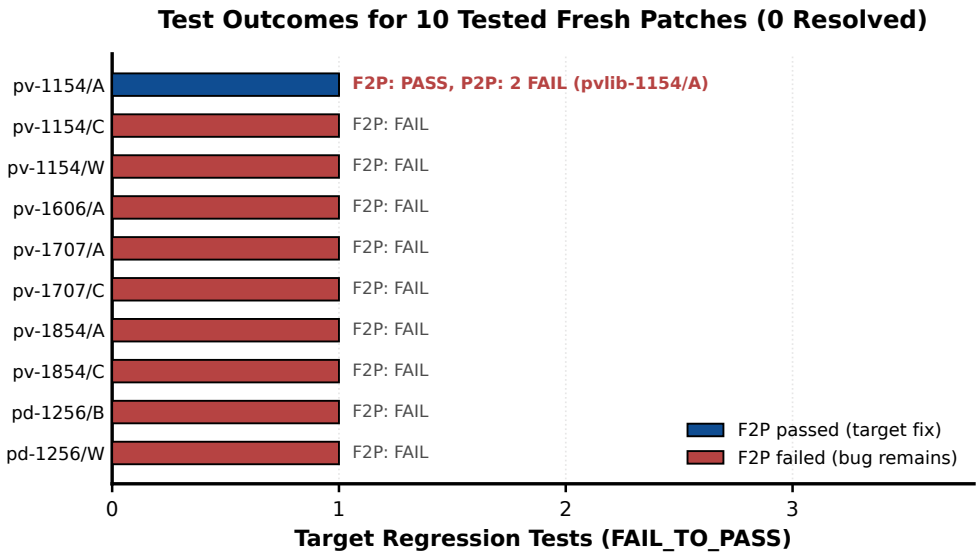



Figure 7: Required test results for the 10 tested fresh patches. Nine patches failed their target regression test (F2P). Only `pvlib-1154/A` passed F2P, but failed two required preservation tests.

syntactically valid patches targeting other visible files, adding zero-division checks to unrelated functions.

### 5.3 Case Study: Generation cap truncation on pydicom

In `pydicom-1256/A` and `pydicom-1413/B`, the model generated repetitive unified diff hunks until reaching the 2,048-token cap. Because generation halted mid-stream, neither output closed its markdown diff fence. The parser recorded "No patch block found," resulting in empty extracted diffs. These failures were caused by the generation reaching the output cap during repetitive output, which prevented the model from emitting the closing markdown diff fence.

## 6 Limitations

Several limitations qualify our findings:

1. **Sample size and sequential bias:** The fresh comparison recovered 22 of 40 attempts across 6 tasks, with complete sets on only 5 tasks. Due to alphabetical task execution, the completed sample is heavily skewed toward `pvlib` (4/5 complete tasks), while `astroid` was unstarted. If additional GPU compute becomes available, completing the remaining 18 runs in randomized execution order is the single most persuasive experimental extension.
2. **Single-attempt decoding:** All runs evaluated greedy decoding (temperature 0, one beam) with exactly one attempt per pair. We cannot estimate variance across stochastic sampling seeds.
3. **Incomplete native controls:** Four of ten native reference controls exhibited skipped or errored preservation tests. While model test failures remain directly observed, positive benchmark resolution cannot be rigorously validated for those four tasks.
4. **Evaluator differences:** The development pilot (Docker SWE-bench 4.1.0) and fresh comparison (native Colab Python 3.9) used different harnesses, prompts, and extraction parsers,

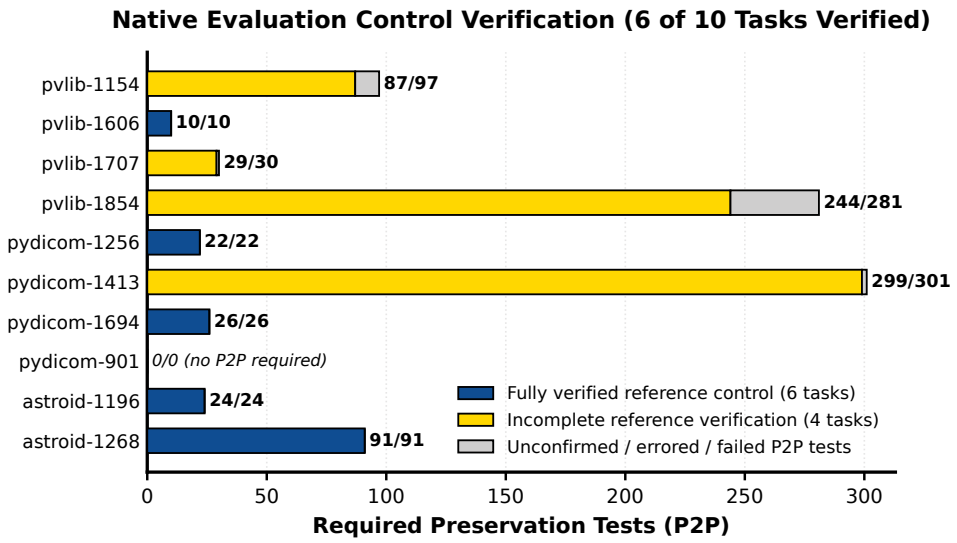



Figure 8: Native evaluation control audit across the 10 fresh tasks. Green indicates fully verified controls (6 tasks); orange indicates tasks with unconfirmed, errored, or failing reference tests (4 tasks).

preventing direct statistical aggregation.

## 7 Reproducibility Statement

All experimental artifacts, frozen evidence packets, checkpoints, and logs are archived with SHA-256 verification. An updated evidence ledger (`evidence_ledger.csv`) maps all numerical assertions to machine-readable records. All tables are reproducible via `python3 paper_v2/build_tables.py`, the reference-coverage diagnostic via `python3 paper_v2/build_reference_coverage.py`, and all figures via `python3 paper_v2/build_plots.py`. No new model inferences were executed during this analysis.

## 8 Conclusion

Under a 4,000-token evidence budget, repository code packaging creates a direct trade-off between file breadth and line depth:

- Whole-file packaging (W) omits reference files in 80% of tasks due to budget exhaustion (displaying 2/10 files and enclosing entities, and 33/55 pooled lines).
- Compact excerpts (B and C) cover reference files (100%) and entities (70%), but truncate the actual edit site in 80–90% of tasks (covering 5/55 and 3/55 pooled lines).
- Complete entity packaging (A) preserves complete bodies for selected functions and classes, exposing edit lines in 3/10 tasks (29/55 pooled lines, 52.7%; per-task mean: 30.0%).
- Across all four policies, the developer edit site is missed in at least 70% of tasks.

Downstream, strict diff application succeeded in only 2 of 22 fresh completions (9.1%), while hunk recounting recovered an additional 8 applicable patches (45.5%). All tested patches yielded zero benchmark resolutions across both cohorts. When the model cannot see the target lines, attempted edits fail due to source-context mismatches in 72.7% of failures, while patches that

apply cleanly modify off-target code. Even when the edit site is visible, fixing a target regression can simultaneously break API preservation contracts. Whether multi-turn execution feedback or adaptive context compression can overcome these structural visibility limits remains an open empirical question for future repository repair benchmarks.

# A Artifact Provenance and Audit Ledger

**Dataset and model revisions.** The SWE-bench Lite development split is pinned to commit `6ec7bb89b9342f664a54a6e0a6ea6501d3437cc2`. The model evaluated is `google/gemma-4-E4B-it` at revision `4ecc2ba63bcac23ba4a767fc87025b1450f84c9c`.

**Archive verification.** The archive manifest (`manifest.json`) lists 311 files, all matching their recorded SHA-256 digests. Local copies of the pilot execution bundles and preflight runs were authenticated against `evaluation_evidence.sha256`.

**Evaluator handoff reconciliation.** The archive contained two versions of `evaluator_handoff/`: an earlier bundle (16:35 UTC, 21 predictions) and a Google Drive handoff (16:41 UTC, 22 predictions). The Drive copy is authoritative, containing the additional completed checkpoint `pydicom-1413/B`.

# References


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